Pardhu Satya Sai Krishna Alapati

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Summary

AI/ML Engineer with 4+ years designing and deploying **production-grade machine learning** systems across clinical and financial domains spanning **transformer fine-tuning**, **LLM**-powered **Retrieval-Augmented Generation (RAG)** and **AI agent** architectures, tabular risk modeling, and end-to-end **MLOps**. Builds and owns the full model lifecycle: from **feature store** design and **deep learning** development through **canary deployment**, **data drift detection**, and regulatory compliance. Applies both large language models (**GPT-4**, **Llama 2**, **BioBERT**) and classical **ML** ensembles (**XGBoost**, **LightGBM**) to high-stakes **production** classification and risk scoring problems. Core stack: **PyTorch**, **LangChain**, **Hugging Face Transformers**, **Azure Machine Learning (Azure ML)**, and **AWS SageMaker**.

Experience

Cigna Healthcare Feb. 2026 – Present

AI/ML Engineer *AL, USA*

- Engineered a **Retrieval-Augmented Generation (RAG)** clinical documentation assistant using **LangChain**, **GPT-4**, and **Pinecone** (**vector database**), cutting prior authorization processing time from 2 hours to 30 minutes across 800 daily prior authorization requests.
- Architected an end-to-end **MLOps pipeline** on **Azure Machine Learning (Azure ML)** with automated retraining triggers, **model versioning**, **CI/CD integration**, and **data drift detection**, compressing model deployment cycles from 3 weeks to 4 days across 4 **production ML** models.
- Fine-tuned a **BioBERT transformer** model on 65,000 de-identified clinical notes for **ICD-10 code prediction** via **Natural Language Processing (NLP)**, achieving a top-3 accuracy of 0.914 on the held-out test set and eliminating 3 hours of manual medical coding daily.
- Launched a **nightly batch inference** pipeline for member risk stratification using **XGBoost** on **AWS SageMaker Batch Transform**, scoring 22,000 member records each night and surfacing 950 high-risk members for proactive clinical outreach.
- Designed a centralized **feature store** with **Apache Feast** and **Databricks Delta Lake**, standardizing 55 reusable features across 5 **production ML** models and removing 4 hours of redundant **feature engineering** per sprint.
- Orchestrated a multi-step **LangChain AI agent** framework with **GPT-4 tool-calling** to automate 3 member inquiry workflows, handling 750 daily member interactions and reducing live-agent escalations from 320 to 95 per week.
- Collaborated with clinical informatics and legal teams to embed **HIPAA compliance** requirements as model constraints and **data governance** policies, advancing **responsible AI** adoption and ensuring all pipelines passed regulatory audit with zero findings.

NVL Lab, University of Alabama Jun. 2025 – Jan. 2026

Graduate Researcher *AL, USA*

- Built a **Python**-based multimodal **data processing** pipeline for the **HoloBMI** project, ingesting and standardizing microscopy imaging, camera video, and behavioral sensor datasets from 3 concurrent recording systems into a unified analysis-ready schema.
- Developed a **timestamp synchronization** module using linear interpolation and **pandas** resampling to align heterogeneous data streams captured at differing sampling rates, achieving sub-10ms temporal precision across all modalities.
- Integrated microscopy, camera, and behavioral datasets into a consolidated **data pipeline** using **NumPy** and **OpenCV**, automating frame extraction, signal segmentation, and format normalization to reduce per-session manual curation from 5 hours to under 1 hour.
- Collaborated with neuroscience researchers to validate cross-modal data alignment and produce structured datasets supporting downstream **statistical analysis** and experimental reporting for the HoloBMI research program.

Morgan Stanley May 2021 – Dec. 2023

ML Engineer *India*

- Developed an ensemble **credit risk model** combining **XGBoost**, **LightGBM**, and logistic regression on 4 years of credit exposure and lending portfolio data, lifting default-prediction **AUC** from 0.71 to 0.84 across $180M in annual credit exposure.

- Built an **anomaly detection** pipeline using **Isolation Forest** and **Keras LSTM** autoencoders to surface suspicious trading patterns, generating 140 compliance alerts per quarter and reducing manual surveillance review by 5 hours weekly.
- Trained a **BERT**-based **NLP** text classifier via **Hugging Face Transformers** on 22,000 analyst research reports for automated **sentiment analysis**, achieving a weighted **F1 score** of 0.89 on the held-out test set, processing 25 weekly publications in under 3 minutes and eliminating 1 analyst-hour of daily manual annotation.
- Optimized **hyperparameter tuning** workflows with **Optuna** and **MLflow** experiment tracking, shortening **model iteration** cycles from 5 days to 18 hours for 3 equities risk models in **production ML**.
- Engineered a 3-class **market regime classifier** using **Random Forest** on yield curve slope, credit spreads, and price momentum features to tag low, medium, and high-risk market conditions across 6 years of S&P 500 data, achieving macro-averaged **precision** of 0.83 and **recall** of 0.80, enabling the risk desk to dynamically recalibrate model thresholds during regime transitions.
- Implemented a **real-time model monitoring** framework with **Apache Kafka** and custom **statistical drift detection**, triggering automated retraining alerts when prediction distributions exceeded defined thresholds on 3 live models.
- Analyzed client portfolio rebalancing patterns across 1,800 accounts using **SQL** and **pandas**, uncovering $18M in underutilized fixed-income allocation opportunities presented to the senior advisory team.
- Migrated 4 years of transaction history from Oracle to an **Apache Spark**-based **data lakehouse** on **Amazon Web Services (AWS)** S3, reducing analytical query execution time from 28 minutes to 6 minutes for the risk analytics team.

Technical Skills

Programming Languages: Python, SQL, Scala, R, Bash
Machine Learning & Deep Learning: PyTorch, TensorFlow, scikit-learn, XGBoost, LightGBM, Keras, Hugging Face Transformers, Deep Learning, Transfer Learning, Model Quantization, Inference Optimization, NLTK, spaCy, OpenCV
Generative AI & LLMs: LangChain, LlamaIndex, AI Agents, Retrieval-Augmented Generation (RAG), Parameter-Efficient Fine-Tuning (PEFT), QLoRA, LoRA, OpenAI API, GPT-4 Tool Calling, BERT, BioBERT, GPT-4, Llama 2, Mistral, Pinecone, Weaviate, ChromaDB, Prompt Engineering
MLOps & Model Governance: MLflow, Apache Airflow, Kubeflow, Optuna, Apache Feast, Databricks, Docker, Kubernetes (K8s), CI/CD Pipelines, Model Governance, Responsible AI, ONNX, TensorRT
Cloud Platforms: Amazon Web Services (AWS), AWS SageMaker, Azure Machine Learning (Azure ML), Google Cloud Platform (GCP), AWS S3, AWS Lambda, Azure Blob Storage
Data Engineering: Apache Spark, Apache Kafka, Delta Lake, Pandas, NumPy, PostgreSQL, MongoDB, Oracle, Apache Hive
Data Analysis & Visualization: Tableau, Power BI, Matplotlib, Seaborn, Plotly, Microsoft Excel, Statistical Analysis, A/B Testing, Hypothesis Testing, Exploratory Data Analysis (EDA)
AI Tools: GitHub Copilot, Cursor, Weights & Biases (W&B), LangSmith, Gradio, Streamlit, Hugging Face Hub, Ollama, Label Studio, Comet ML
Tools & Practices: Git, FastAPI, Jupyter Notebooks, REST APIs, CUDA, Mixed-Precision Training (AMP), HIPAA Compliance, Data Governance, Responsible AI, Feature Engineering, System Design

Projects

Clinical Q&A System via Parameter-Efficient LLM Fine-Tuning

- Adapted **Llama 2 (13B)** to clinical question answering using **QLoRA** (**Parameter-Efficient Fine-Tuning / PEFT**) on 12,000 MedQA and PubMedQA pairs, achieving an **accuracy** of 0.78 on the held-out evaluation set — 14 points above the few-shot prompting baseline.
- Applied **model quantization** via **GPTQ** to 4-bit precision, cutting **inference** memory from 26 GB to 7 GB, then packaged it as a **Gradio** application published to **Hugging Face Hub** for zero-infrastructure researcher access.

Multi-GPU Distributed Training Pipeline

- Deployed a **distributed training** setup using **PyTorch Distributed Data Parallel (DDP)** across 4 **GPU**-backed AWS EC2 p3.2xlarge instances, reducing transformer model training time from 10 hours to 3 hours.
- Integrated **gradient checkpointing** and **automatic mixed-precision (AMP)** training, lowering per-GPU memory from 14 GB to 8 GB on each V100, then containerized the environment with **Docker** and orchestrated job scheduling via **Kubernetes (K8s)** for reproducible multi-node experiments.

Education

University of Alabama at Birmingham — Birmingham, AL, USA
Master of Science in Computer Science — *Jan. 2024 – Dec. 2025*

KKR & KSR Institute of Technology and Sciences — Guntur, India
Bachelor of Technology in Computer Science — *Apr. 2019 – Jun. 2023*